

RETAIL SALES DATA ANALYSIS & RFM SEGMENTATION REPORT

A comprehensive analysis of retail sales transactions to understand customer purchasing behavior, response patterns, and revenue contribution using RFM segmentation.

INFORMATION

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Role: Data Analytics Intern

Tools Used: MySQL | Python (Pandas, Matplotlib) | Excel

Project Focus: Customer Segmentation & Business Insights

Date: February 2026

Executive Summary

This project analyzes retail sales data to evaluate customer behavior and campaign performance.

A total of **6,884 customers** and **125,000 transactions** were analyzed. Using the RFM (Recency, Frequency, Monetary) model, customers were segmented into:

- Champions
- Loyal Customers
- Frequent Customers
- Recent Customers
- Others

Key Findings:

- 647 customers** responded to the campaign (**9.4% response rate**)
- Responders spend more on average than non-responders
- Champions and Loyal customers generate higher revenue

Tools Used:

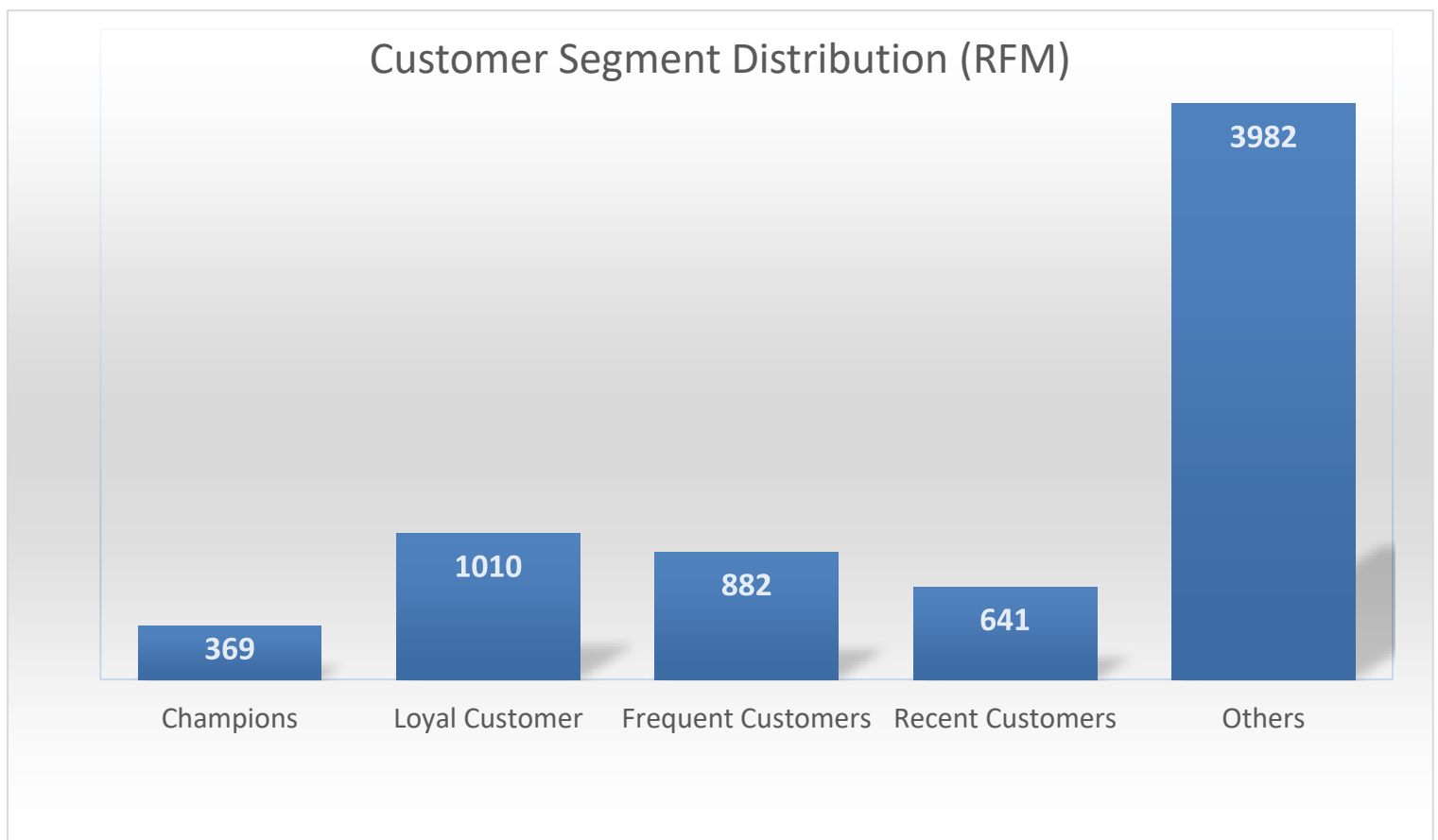
- MySQL (Data extraction & aggregation)
- Python (Data cleaning & RFM analysis)
- Excel (Reporting & visualization)

Business Recommendation

Prioritize Champions and Loyal Customers

- Target “Others” to increase customer lifetime value
- Use RFM-based personalization
- Continuously monitor campaign ROI

Customer Segment Distribution (RFM Analysis)



Key Insights

- 3,982 customers fall under the “Others” segment (largest group)
- Loyal Customers (1,010) represent strong repeat buyers
- Champions (369) contribute disproportionately high revenue
- Opportunity exists to convert Frequent & Recent customers into higher-value segments

THANK YOU

Presented By

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Tools Used: MySQL | Python | Excel

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